

JEFFRIN RIJO V C

Unreal Engine C++ Game Dev | Software Engineer | Technical 3D Artist

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Portfolio: <https://jeff3joo.netlify.app>



Unreal Engine C++ Game Developer focused on gameplay mechanics and real-time systems. Strong software engineering background with experience in building interactive systems and integrating game-ready assets using Blender.

EXPERIENCE

VisAI Labs

Full Stack Intern (Feb 2024 – Dec 2024)

Full Stack Engineer Trainee (Jan 2025 – Present)

- Worked on **vMeasure-CodeScan** barcode scanning product, improving measurement **accuracy, reliability,** and **processing speed.**
- Implemented **serial port communication** for stable device-to-system data exchange.
- Contributed to a high-performance **C++ SDK wrapper** for barcode detection via Scandit Barcode Reader and Zebra AVL enabling efficient real-time capture from camera streams.
- Integrated multiple **barcode SDKs**, tested accuracy, and automated data validation flows.
- Enhanced a Chrome Extension to trigger automated barcode actions through API via **vMeasure/vAudit Forge.**

Alhansat Solutions

Full Stack Developer Intern Sep 2023 – Oct 2023 (45 days)

- Co-developed a Flowchart Maker web tool with modular architecture.
- Built drawing features, module interaction, and export pipeline.

PROJECTS

Unreal Engine C++ Projects

Battle Blaster v2 (Unreal Engine C++)

- Designed a reactive enemy AI system with distance-based state transitions (Chase → Attack).
- Implemented dynamic respawn logic with last-known-position tracking.
- Built health reward mechanics tied to enemy death events.
- Developed interruptible movement behavior prioritizing player detection over patrol targets.
- Integrated sound effects, particle systems, and camera shake for feedback.

Obstacle Assault (Unreal Engine C++)

- Developed a physics-based obstacle course game using Unreal Engine C++.
- Implemented Actor movement using Delta Time, velocity, and vector math.
- Built rotating platforms and moving obstacles with collision-based logic.

Dungeon Escape (Unreal Engine C++)

- Created an escape-room style game with trigger-based interactions and puzzles.
- Built input handling using Unreal's Enhanced Input System in C++.
- Used line tracing and collision detection for player interaction mechanics.

Shooter Sam (Unreal Engine C++)

- Developed a third-person shooter with character animation systems and AI behavior.
- Implementing Animation Blueprints, Blend Spaces and state machines.
- Building gun mechanics with line tracing, particle effects, and damage systems.
- Worked with AI Controllers, NavMesh, Behavior Trees, and Blackboards.

3D Art & Animation Projects (Blender)

- Created game-ready props, low-poly environments, and animated assets.
- Worked across modeling, shading, lighting, and rendering pipelines.

- Optimized assets for real-time use and integrated them into Unreal Engine projects.
 - Portfolio: <https://jeff3joo.netlify.app/projects>
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Website Projects

Portfolio (React, MUI)

- Designed and developed a responsive portfolio platform to showcase 3D projects.
- Integrated a 3D model viewer for GLB assets with image and video previews.
- Implemented smooth UI animations and mobile-optimized layouts.
- Live Demo: <https://jeff3joo.netlify.app>

Weather Web App, RPS Game

- Weather Web App using OpenWeatherMap API with responsive UI.
 - Rock-Paper-Scissors browser game with score tracking and animations
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SKILLS

GAME DEVELOPMENT

- Unreal Engine 5 - C++
- Gameplay Programming
- Actor & Component Architecture
- Pawn / Character Systems (Basics)
- Camera & Spring Arm Systems
- Collision & Trigger-Based Mechanics
- Blueprint + C++ Integration

SOFTWARE ENGINEERING

- C++
- Git
- JavaScript
- React JS
- Node JS
- SerialPort Comm.

TECHNICAL ART & DESIGN

- Blender (Game-Ready Asset Creation)
 - Modeling, Shading & Lighting
 - Basic Rigging & Animation
 - Unreal Asset Integration & Optimization
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EDUCATION

B.TECH (AI & Data Science)

Velammal Institute of Technology (2020-2024) **8.47 CGPA**

HSC

Maria Rafols Matriculation Higher Secondary School (2019-2020) : **71%**

SSLC

Maria Rafols Matriculation Higher Secondary School (2017-2018) : **85.8%**

CERTIFICATIONS

- Meta Front-End Developer Professional Certificate (Coursera)
- Developer Foundations, Node JS, React JS (NXT Wave)
- Google UX Design Professional Certificate (Coursera)
- Complete Blender Creator: 3D Modelling (Udemy)
- Unreal Engine 5 C++ Game Development(Udemy)